TANG CHEE HENG

DeFi Protocol Researcher | Smart Contract Engineer

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SUMMARY

Blockchain engineer and DeFi researcher specializing in protocol architecture, yield strategies, and smart contract development. Built security-audited core smart contracts powering Polystream, an ERC-4626 multi-chain yield protocol integrating 35+ DeFi yield strategies. Experienced in DeFi protocol analysis, on-chain yield mechanisms, and building full-stack Web3 products across mobile, web platforms, and decentralized applications, with a background in artificial intelligence and machine learning. Seeking roles in on-chain research and analysis, smart contract engineering and yield protocol infrastructure.

PROFESSIONAL EXPERIENCE

Co-founder, Protocol Engineer, Polystream **Feb 2025 - Present**

- Co-founded Polystream, a DeFi yield aggregation protocol originating from the Scroll Open Hackathon (2nd place, $15K USD in prize), selected for the Scroll Campus accelerator program, and later securing $95k USD in Scroll Foundation investments and grants.
- Designed and built the core Solidity smart contracts powering an ERC-4626 vault architecture supporting multiple assets across 3 blockchain networks.
- Researched and evaluated DeFi protocols and on-chain yield strategies across lending markets, liquidity pools, yield tokenization, leveraged yield looping, cross-chain liquidity, and stablecoin-based yield mechanisms, reviewing protocol architectures and smart contracts to assess integration feasibility, yield sustainability, capital flows and security considerations.
- Integrated 35+ DeFi protocols and yield strategies such as Pendle, Aave, Curve, Balancer and Morpho into a cross-chain vault architecture, engineering allocation mechanisms that allows capital distribution across strategies to target 8–12% APY.
- Implemented safeguarded curator-controlled strategy execution with whitelisted and guarded smart contract interactions, ensuring capital security.
- Developed Foundry-based testing infrastructure including unit, integration tests and fuzz testing covering vault accounting, strategy integrations, and edge-case validation.
- Built backend infrastructure including event listeners, oracle integrations, monitoring services, and vault trackers for real-time vault management.
- Built full-stack applications including a web dApp, mobile DeFi app, and curator dashboard for vault monitoring and strategy management.
- Led the security audit process with Cantina for Polystream's core smart contracts, collaborating with auditors to explain protocol architecture, discuss findings and implementing fixes.

Fullstack AI Intern, Vilor Berhad **June 2024 - Sep 2024**

- Developed the Vilor internal intranet platform, enabling centralized document management, HR workflows, and operational tools used across finance, HR, and administration teams.
- Implemented HR process automation including dynamic form-to-document generation (Word, Excel, PDF) supporting 19 HR workflow templates and 31 job description templates, improving HR workflow efficiency by 20% and reducing document generation errors by 70%.
- Designed internal systems for document storage, form submission, and company-wide information management, including fiscal calendars, company holidays, and centralized reference resources for employees.
- Processed and analyzed 13 million WhatsApp sales advisory messages using Python, extracting 2,770 frequently asked customer questions to construct a training dataset and knowledge base for onboarding new sales staff.
- Built an AI chatbot prototype for sales and customer service training by fine-tuning a large language model on constructed knowledge base derived from internal communication data.

PROJECTS

Decentralized Credit Scoring for DeFi Lending, Final Year Project

- Analyzed 2,000+ on-chain wallets, extracting behavioral features from transactions, asset holdings, and smart contract interactions to construct a DeFi credit scoring dataset.
- Trained CatBoost, Gradient Boosting, and Random Forest models, achieving ~98% prediction accuracy (R^2 up to 0.985) for wallet credit risk assessment.
- Built a semi-supervised learning pipeline in Python and integrated Chainlink External Adapters + Chainlink Node to submit and store credit scores on-chain via smart contracts.
- Developed a full-stack Web3 dApp allowing users to submit wallet addresses and retrieve generated credit scores

DeESG – Decentralized ESG Scoring Platform, ETH GLOBAL BANGKOK

- Winner – Chainlink "Connect the World" ($2K), Innovation on Scroll 5th Place ($500).
- Built a decentralized ESG scoring platform using AI clustering and IoT environmental data from Raspberry Pi sensors.
- Implemented Chainlink oracles, external adapters, and Chainlink Automation to push verified IoT data on-chain for ESG evaluation.
- Developed smart contracts and token incentives to store ESG scores and reward verified environmental data, integrating FHE-based anonymous governance forums.

AI Guardian – Deepfake & Image Verification, ETH KL

- Winner – ETH KL Best Community Application on Scroll ($1K), ICP–Ethereum Chain Fusion Challenge ($1K).
- Built an AI-powered deepfake detection platform with on-chain verification using ICP Canisters and World ID.
- Developed smart contracts to store image hashes and classify images as authentic or deepfake.
- Integrated AI inference, blockchain storage, and frontend systems for real-time verification.

TECHNICAL SKILLS

Blockchain & Smart Contracts

Solidity, Foundry (Unit, Integration, Invariant & Fuzz Testing), Hardhat, Ethers.js, Wagmi, Chainlink (Data Feeds, Automation, Data Streams, Node), Pyth, RedStone, Account Abstraction, Smart Accounts, Bundlers, Paymasters

DeFi & On-chain Research Tools

DeFiLlama, Dune Analytics, DeBank, Etherscan, Tenderly, The Graph, Moralis, Infura, Alchemy

Backend & APIs

Node.js, Express, FastAPI, Docker

AI / Machine Learning

Python, Scikit-learn, CatBoost, XGBoost, K-Means, FAISS, LangChain, HuggingFace, OpenAI

Frontend

React, Next.js, Vue.js, React Native, TypeScript, JavaScript, Tailwind CSS

Databases

PostgreSQL, TimescaleDB, MongoDB, Firebase, SQL

EDUCATION

2021-2025

University of Nottingham

BSc Computer Science with Artificial Intelligence

QS World Ranking: Top 100
First Class Honours with average result of 79%
Dean's List Scholarship Award Top 2%(2022, 2024)

SMJK JIT SIN

SPM: 10A (5A+, 4A, 1A-)